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TECHNOLOGICAL
UNIVERSITY**
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**College of Computing and
Data Science**

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 5 – Sensor Fusion: Complementary Filter

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Recap: Accelerometer Operation

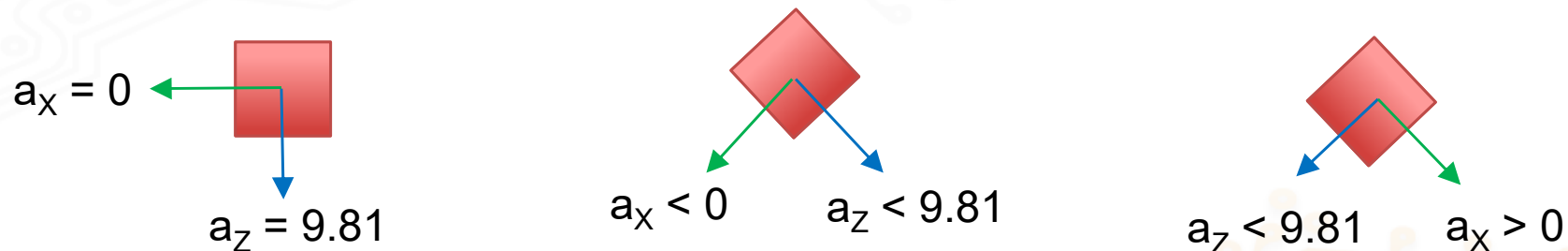
Accelerator measure acceleration along its 3 axes (x, y, z)

- when stationary and lying flat on the surface w.r.t. to earth
- only a_z detects the gravity acceleration: 9.81 m/s^2

When tilted (but remain stationary)

- gravity acceleration measured by a_z will be less
- while the other axes will detect some amount of gravity acceleration

Example: a 2-Axis measurement (tilted along the Y-axis)



Note that a_z measures less, but measure same amount of acceleration when tilted the same angles in either directions.

Recap: Accelerometer Characteristics

Accelerometer measures acceleration

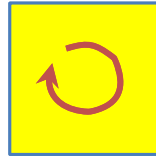
- provides accurate results when it is stationary or moving under non-accelerating condition
- susceptible to disturbances such as rotating, lateral and vibrating movements that induces acceleration
 - can applying a low pass filter to reduce/remove the readings due to these disturbance

Accelerator is hence a good indicator to measure pitch and roll orientations using the gravity vectors

- when in static condition (i.e., **long term** steady state)

Recap: Gyroscope Operation

Gyroscope measures the rate of rotation along its 3 axes



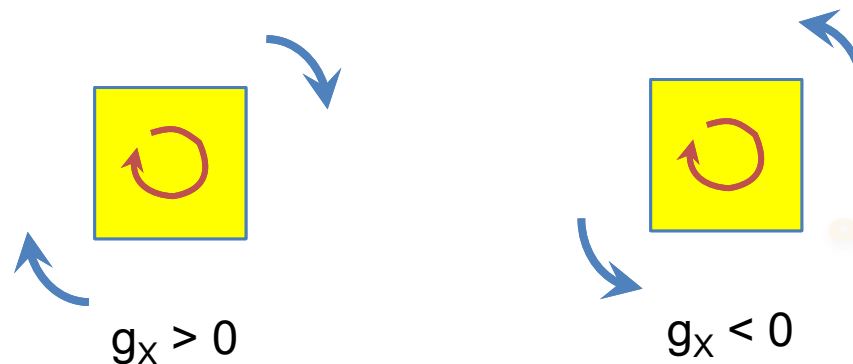
When stationary

- all 3 axes g_x , g_y , g_z measure zero values

When rotating

- values measured can be +ve or -ve, depending on direction of rotations

Examples:



Recap: Gyroscope Characteristics

Gyroscope gives accurate values when it detects rotating movement

- not susceptible to other external forces caused by other disturbances
- Pitch, Roll and Yaw angles can be derived by integrating its output values

But Integration is susceptible to drift in the data over time

- occurs in gyroscope even when there is no rotating movement

Drift is a slow changing value

- can be removed by using a high pass filter
 - without affecting its short term data readings

Sensor Fusion Concept

Sensor fusion combines data from different sensors

- to derive more accurate value by overcoming the limitation of individual sensor

For measurements of Pitch and Roll angle using IMU

- Accelerator can provide accurate result by passing its readings through a low pass filter
 - but degrade its short term dynamic performance
- Gyroscope can provide accurate result over short time interval under dynamic condition
 - its long term DC drift effect can be removed by passing its data through a high pass filter

Sensor Fusion Techniques

Hence we can apply Sensor Fusion techniques that combines the good performance of each sensor

- Long term stability and accuracy of accelerometer
- Short term fast dynamic response of the gyroscope

Two common Sensor Fusion techniques used to detect Pitch and Roll angles from IMU sensors

- i. Complementary Filter
- ii. Kalman Filter

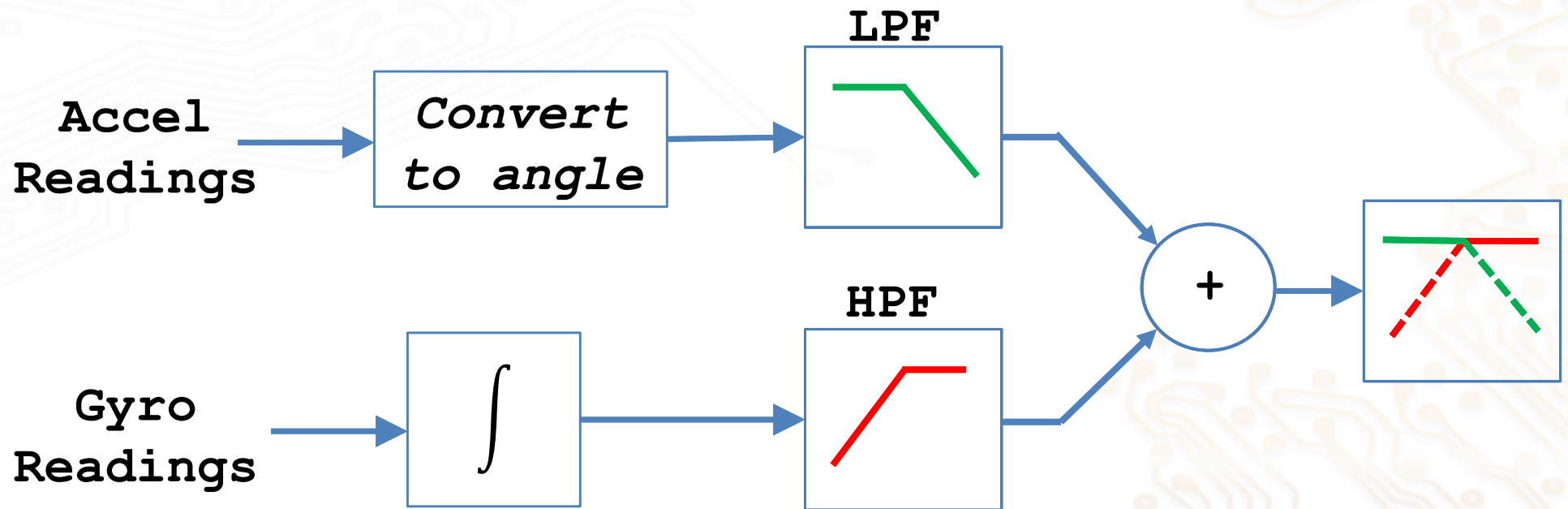
We will first study the implementation of the Complimentary filter to calculate the Pitch and Roll angles

- relatively easy to implement digitally

Complementary Filter Concept

A complementary filter consists of two filters

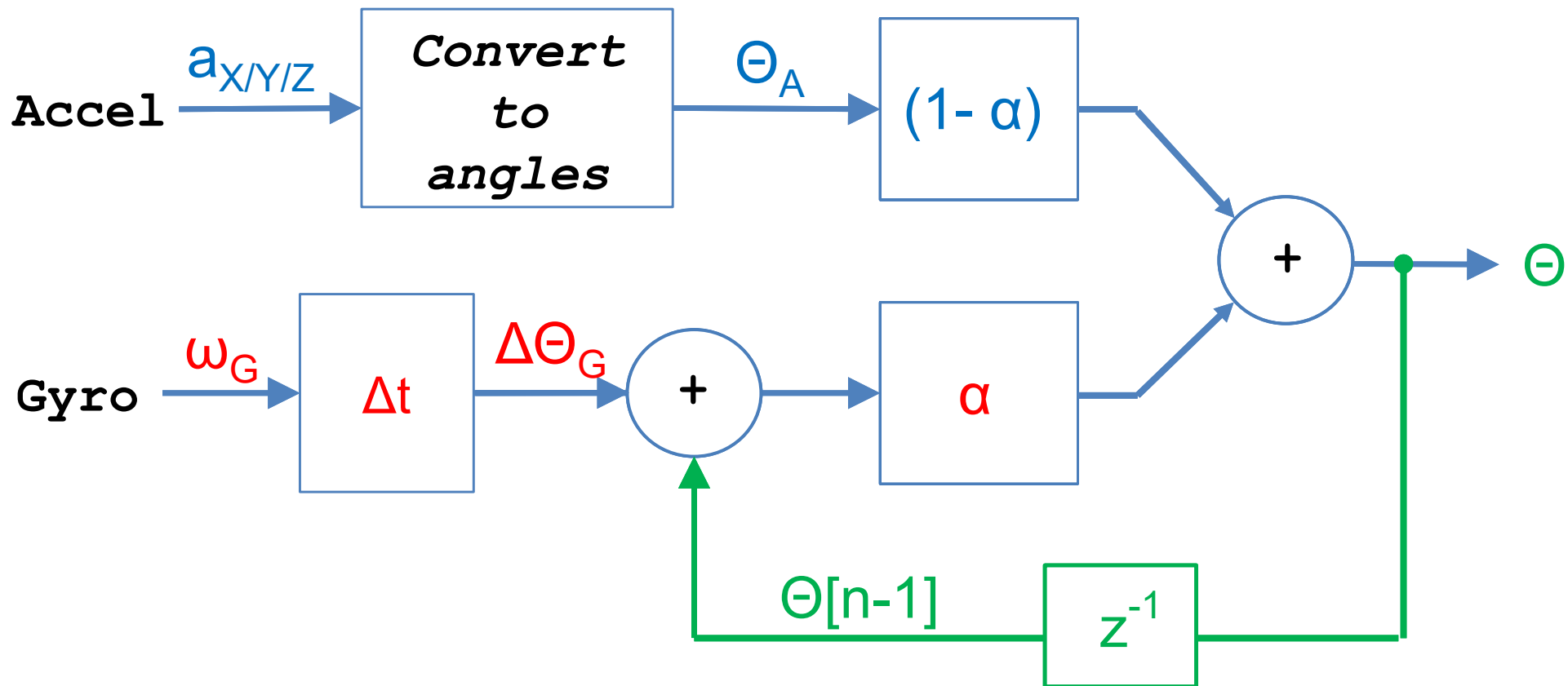
- a low pass filter and a high pass filter
 - that complement each other
- output is the sum of the filters output
 - of different proportion (weightage)



Discrete Time Complementary Filter

Complementary filter can be implemented in discrete time domain using the general equation

$$\Theta[n] = (1 - \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$



Discrete Time Complementary Filter (cont.)

General equation of Complementary Filter:

$$\Theta[n] = (1 - \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$

Q: Where are the 'LPF' and 'HPF'?

(a) Consider the case when the gyroscope reading

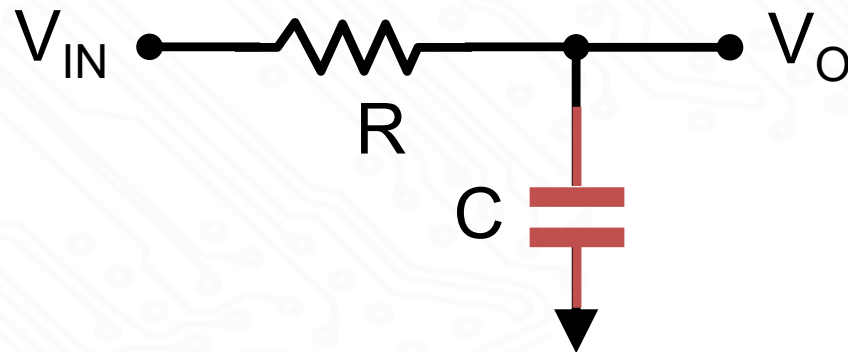
$$\omega_G = 0$$

$$\Theta[n] = (1 - \alpha) * \Theta_A[n] + \alpha * \Theta[n-1]$$

This has the general form of the Low Pass filter equation we previously derived (see next slide):

$$V_O[n] = (1 - \alpha) * V_{IN}[n] + \alpha * V_O[n-1]$$

Recap: Digital Low Pass Filtering



Current flowing through R:

$$I_R = (V_{IN} - V_O)/R$$

Current flowing through C:

$$I_C = C \, dV_O/dt$$

$$I_R = I_C \Rightarrow V_O + RC \, dV_O/dt = V_{IN} \quad \text{----- (1)}$$

$$dV_O/dt = (V_O[n] - V_O[n-1]) / T \quad \text{----- (2)}$$

Substitute (2) into (1) and rearranging:

$$V_O[n] = T/(T+RC) \, V_{IN}[n] + RC/(T+RC) \, V_O[n-1]$$

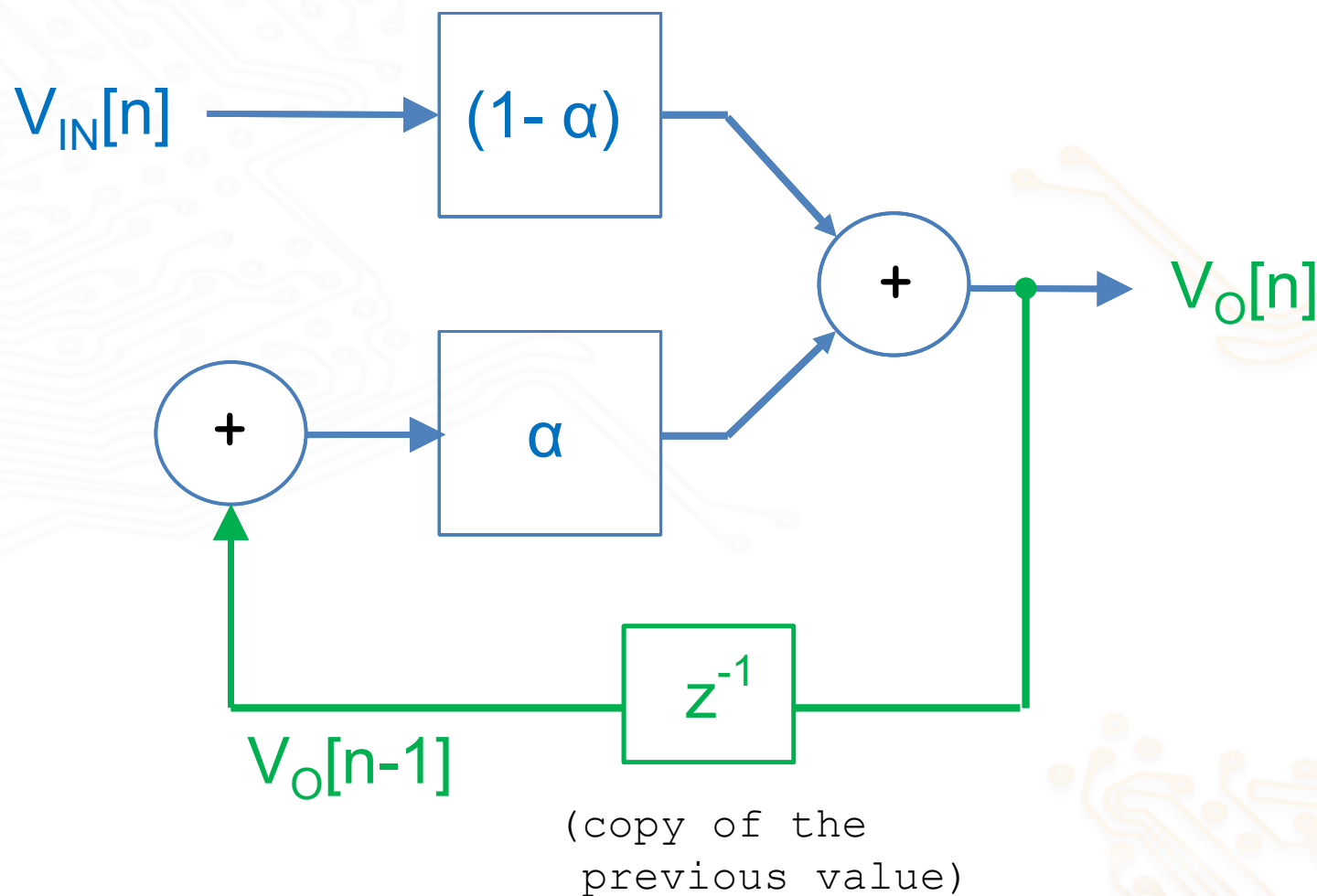
Note that $T/(T+RC) + RC/(T+RC) = 1$

Setting $RC/(T+RC) = \alpha$

- $V_O[n] = (1 - \alpha) \, V_{IN}[n] + \alpha \, V_O[n-1]$

Discrete Time LP Filter Implementation

$$V_O[n] = (1 - \alpha) V_{IN}[n] + \alpha V_O[n-1]$$



Discrete Time Complementary Filter (cont.)

General equation of Complementary Filter:

$$\Theta[n] = (1 - \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$

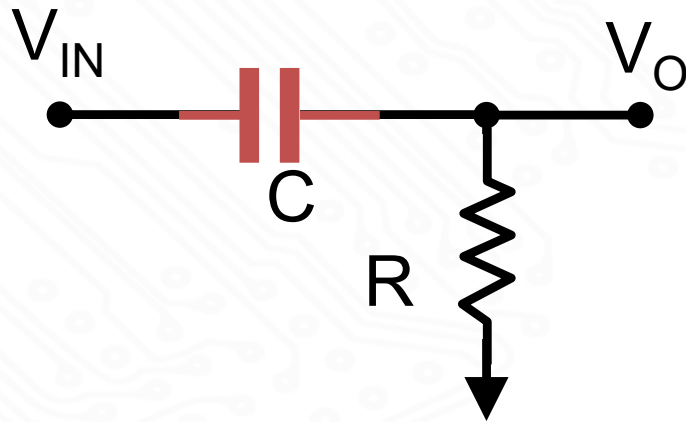
(b) Consider the case when the Accelerometer derived angle $\Theta_A = 0$

$$\begin{aligned}\Theta[n] &= \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t) \\ &= \alpha * (\Theta[n-1] + (d\Theta_G / \Delta t) * \Delta t) \\ &= \alpha * \Theta[n-1] + \alpha * d\Theta_G \\ &= \alpha * \Theta[n-1] + \alpha * (\Theta_G[n] - \Theta_G[n-1])\end{aligned}$$

This has the general form of the High Pass filter equation (see next slide):

$$V_O[n] = \beta V_O[n-1] + \beta (V_{IN}[n] - V_{IN}[n-1])$$

Recap: Digital High Pass Filtering



Current flowing through R :

$$I_R = V_O / R$$

Current flowing through C :

$$I_C = C \, d(V_{IN} - V_O) / dt$$

$$I_R = I_C \quad \Rightarrow \quad V_O / R = C \, d(V_{IN} - V_O) / dt$$

$$\text{Rearranging} \quad V_O + RC \, dV_O / dt = RC \, dV_{IN} / dt$$

Implement differentiation digitally in software:

$$V_O[n] + RC(V_O[n] - V_O[n-1]) / T = RC(V_{IN}[n] - V_{IN}[n-1]) / T$$

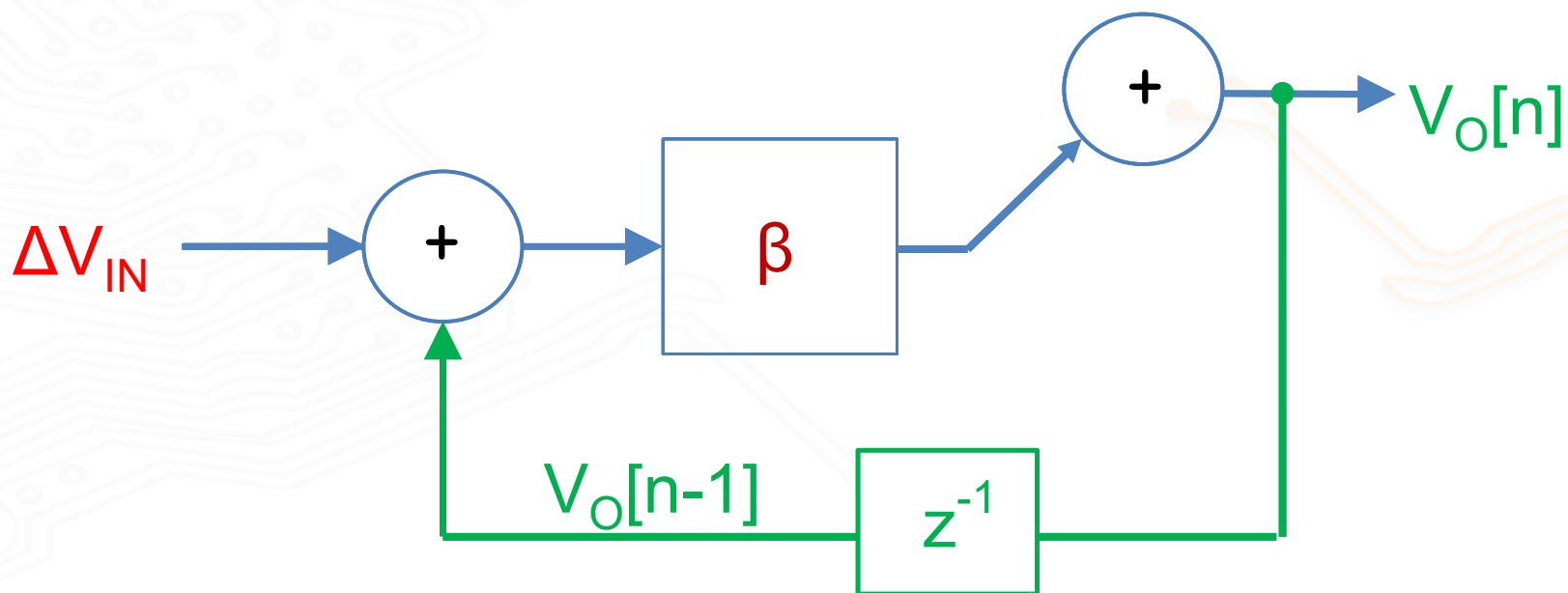
Rearranging:

$$V_O[n] = RC / (T + RC) \, V_O[n-1] + RC / (T + RC) \, (V_{IN}[n] - V_{IN}[n-1])$$

$$\text{i.e. } V_O[n] = \beta \, V_O[n-1] + \beta \, (V_{IN}[n] - V_{IN}[n-1])$$

Discrete Time HP Filter Implementation

$$V_O[n] = \beta V_O[n-1] + \beta (V_{IN}[n] - V_{IN}[n-1])$$



Choice of the Weightage Factor α value

$$\Theta[n] = (1 - \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$

As the α value is associated with the output of the gyroscope

- it is set to a large value
 - E.g., $\alpha = 0.7$ to 0.9
- such that the gyroscope has a large but short term effect

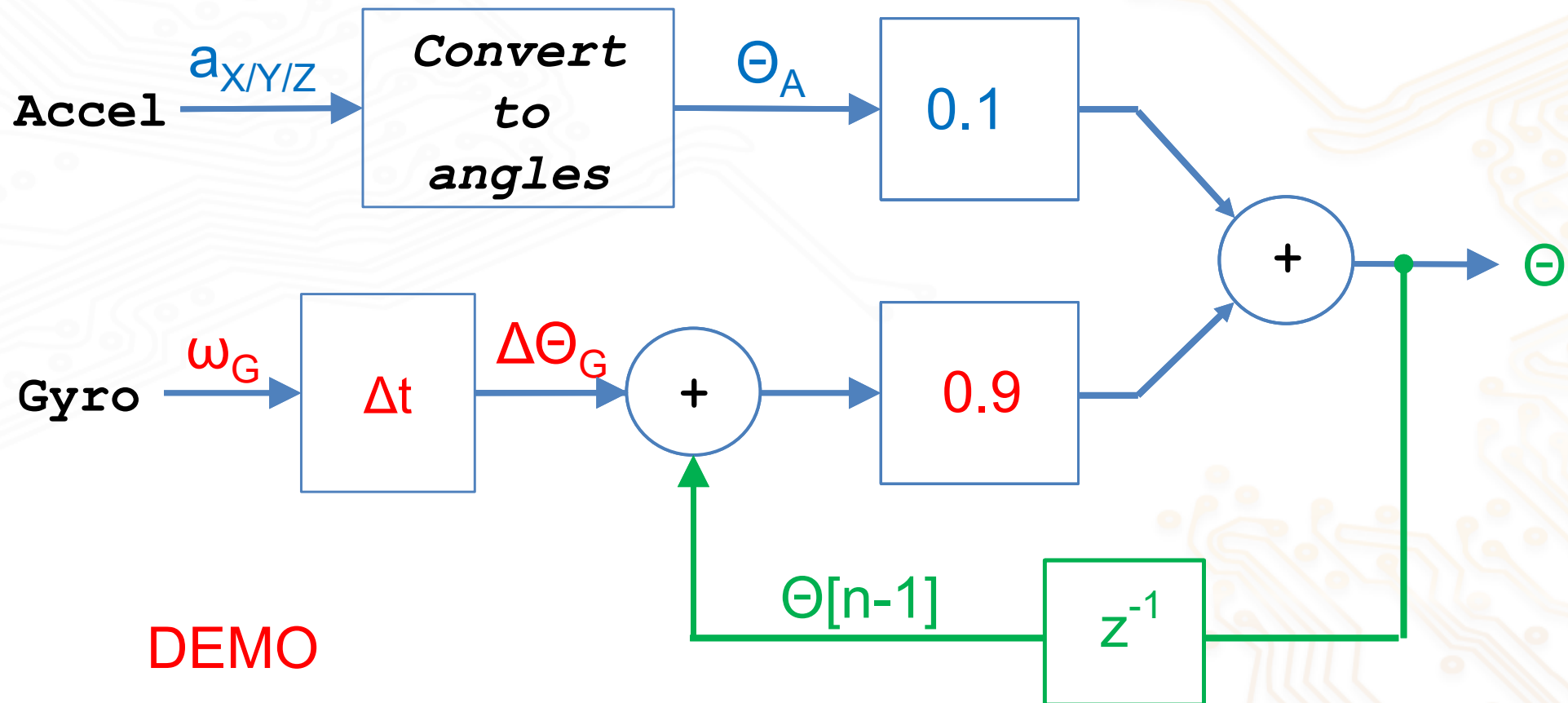
The weightage left for the accelerometer's derived angles is hence

- $(1 - \alpha) = 0.3$ to 0.1

Discrete Time Complementary Filter Implementation

Using $\alpha = 0.9$ (i.e. for the Gyroscope)

$$\Theta[n] = 0.1 * (\Theta_A[n]) + 0.9 * (\Theta[n-1] + \omega_G[n] * \Delta t)$$



Summary

Sensor Fusion technique

- combines data from different sensors
- able to produce more precise results

Complementary Filter

- uses different filter types that target the limitations of each sensor in IMU (i.e., complement each other performance)
- able to produce more responsive and more stable performance when measuring the Pitch and Roll angles

But the weightages used in CF are largely based on intuition or empirical approach

- how do we find the optimum value?

A more scientific and better performance method: **Kalman filter**

- where the weightage is adjusted continuously for optimum settings.